

Container Management System - Scrum Sprint Documentation (Sprint1)

Introduction

This document gives a summary of the Agile Scrum process followed during the development of the Container Management System. The project was managed over 6 two-week sprints, with each sprint delivered a define set of user stories and business objectives. Progress was managed by Trello, while GitHub commit history, implementation milestones and testing evidence were used to verify the completion of sprint deliverables.

Sprint 1

Sprint Goal: Establish the project foundation by completing the initial project planning, system design, backend project structure, user authentication, role-based access control and container registration functionality required for subsequent development.

Sprint Backlog

ID	User Story	Priority	Story Points	Status
PB-01	Project planning and documentation	Must Have	3	Done
PB-02	Establish project structure	Must Have	2	Done
PB-03	User Authentication	Must	3	Done
PB-04	Role-Based Access Control	Must	5	Done
PB-05	Register Container	Must	3	Done

Development Tasks

- Complete the project proposal and requirements analysis
- Produce the Entity Relationship Diagram and initial documentation
- Configure the Express project structure and folders
- Implement JWT-based authentication and secure user login
- Develop the container registration
- Verify authentication and registration using Thunder client and database testing

GitHub Evidence

- 27 June 2026- Initial project setup with proposal, ERD and documentation
- 28 June 2026- correct project structure
- 29 June 2026- Implement JWT authentication and user login
- 29 June 2026- Complete JWT authentication module
- 29 June 2026- Implement container management module

Definition of done

- Sprint objectives achieved
- Business requirements implemented
- Authentication and authorization validated
- Features tested using Thunder client and database verification
- Source code committed to GitHub
- Sprint deliverables integrated successfully

Sprint Review

Sprint 1 successfully established the technical foundation for the Container Management System. The project planning and system design activities were completed before development started. This provided a clear structure for the application. The user authentication, role-based access control and container registration module were successfully implemented and tested. This enables secure access to the system.

What Went Well

- Project objectives and requirements were clearly defined
- A solid backend and folder structure were established
- JWT authentication was successfully implemented
- Secure role-based access control as achieved
- Container registration functionality was completed and tested

Challenges

- Learning and implementing JWT authentication
- Designing an effective database schema and relationships
- Establishing a scalable project structure for future development

Focus for Sprint 2

- Develop the workflow engine
- Implement container movement tracking
- Validate workflow transitions
- Implement QA approval process
- Begin Supervisor approval workflow